

Week 1 — Reading Pack

Curated excerpts from industry white papers, organized by learning objective

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Required reading before Week 2 · ≈35 minutes

Assigned alongside: IQVIA Institute, “The Use of Medicines in the U.S.” (latest annual report — executive summary), and the course glossary.

Why a reading pack. The syllabus lists industry primers and white papers as core readings because the vocabulary of pharma commercial analytics is not in any textbook — it lives in the documents practitioners actually write. Below are verbatim excerpts (quoted, with source) from nine IQVIA white papers in the course library, each chosen because it illustrates one of tonight's learning objectives. Read the excerpts, then answer the reflection questions; bring one answer to Week 2. The full PDFs are in the course site's “White papers” folder if you want to go deeper.

Learning objectives for Week 1 (from the syllabus)

1. Articulate why pharma commercial analytics is structurally different from B2C analytics.
2. Identify the major stakeholders (manufacturer, payer, provider, patient, PBM, GPO, wholesaler) and their economic incentives.
3. Read a basic drug brand P&L and define gross-to-net.

Key concepts: the three Ps (payers, providers, patients) plus policymakers; the flow of a prescription from script to fill to claim to revenue; HCP, HCO, IDN, PBM, ICD-10, NDC, NPI.

A. Why launches — and forecasts — go wrong (objective 1)

The single most useful number to carry into this course: most launches miss their first-year forecast. IQVIA's launch-excellence research explains why, and introduces a four-archetype framework you will meet again in Week 6 (forecasting).

In studying about 12,000 product launches across eight countries and 10 years, one of the most important factors is setting expectations. Between 50% and 70% of launch brands underperform their first-year forecast. That suggests companies routinely create unrealistic expectations or fail to understand the opportunity. [...] The most critical challenge is the launch strategy. Too many companies use the same playbook for every launch. Just as every product is different, every launch strategy and implementation must be unique to the product, market, opportunities and challenges. [...] A brand's fate is sealed before the launch even begins. The quality of pre-launch preparation, resources, alignment and processes determines post-launch success.

— IQVIA, “A Framework for Successful Biopharma Launches” (Gupta & McClellan), Critical Success Factors

IQVIA analyzed more than 600 launches based on two axes of unmet need and product differentiation. What they found was four archetypes with distinct critical success factors and distinct first-year sales. [...] The upper right quadrant (high unmet need and high product differentiation) is labeled Science Sells. Products in this area have the highest average first-year sales: \$94 million. The other end of the spectrum, low unmet need and low product differentiation, is Who Benefits, with average first-year sales of \$19 million. [...] The other archetypes are Market Shaping, average first-year sales of \$65 million for products

with low market need and high product differential, and Emphasize the Difference, \$53 million in first-year sales with high unmet need and low product differentiation.

— IQVIA, “A Framework for Successful Biopharma Launches,” The Framework

Low unmet need and low product differentiation is a tough position for any product. That's when managing expectations becomes crucial. If marketers can't identify the distinct patient population or stakeholder who would benefit, the launch will likely fail. That was the position Amarin found itself in when launching Vascepa. [...] The mechanism of action inspired revenue estimates of \$60 million for 2013 and \$1.9 billion in peak sales for 2022. Prescribers did not agree. From the physician perspective, a statin was all that most patients needed. [...] Vascepa generated only \$32 million in first-year sales, less than half of Lovaza's first-year sales. New prescription results were even lower: 94.2k new scripts for Vascepa versus 402.6k new scripts for Lovaza.

— IQVIA, “A Framework for Successful Biopharma Launches,” Who Benefits case

Reflection A. Which stakeholder did Amarin's forecast implicitly assume would behave differently than they did — the payer, the prescriber or the patient? What data, available before launch, could have tested that assumption?

The economics behind the pressure

Developing novel therapies to address unmet patient need is the lifeblood of the biopharmaceutical industry. It is also a capital-intensive and high-stakes endeavour, requiring an estimated \$3.1 billion to bring a new therapy to market, with a composite success rate of 11% from phase 1 through regulatory submission. Clinical trial readouts, therefore, represent major inflection points in this journey, as moments of truth, that resolve uncertainty around an asset's future prospects, including its ultimate potential for revenue generation.

— IQVIA, “Inflection Point: How Clinical Trial Results Impact Biopharma Valuations” (Gores, Harries, Capuzzi), Introduction

Why it matters for a commercial analyst: by the time a brand reaches you, the company has usually spent a decade and billions of dollars; the exclusivity window in which to recover that investment is typically 8–12 years. Every forecasting and ROI decision you make sits inside that clock.

B. Stakeholders and their incentives (objective 2)

Policymakers and safety-net providers: the 340B program

340B is a good first example of how a single policy creates a stakeholder with its own incentives — and a wedge between list and net price that shows up in every brand P&L.

The 340B drug pricing program (“340B program”) is a federal program in which drugmakers provide substantial, medicaid-like discounts on qualifying drugs to participating hospitals and clinics. The program was designed so that 340B providers would use 340B revenue to help vulnerable individuals access drugs and healthcare. [...] In 2025, 340B purchases reached \$179.5 billion at list prices, growing 20.3% year-over-year. From 2018 to 2025, 340B purchases at list prices more than tripled in size and grew at three times the rate of the rest of the U.S. drug market. Adjusted for missing sales in our sample, we estimate that total program purchases at list prices were approximately \$200 billion, which is approaching the size of medicare Part D.

— IQVIA, “The Size and Growth of the 340B Program in 2025” (Martin, Karne, Zeng), Abstract & Introduction

A recent study published by our group found that 80% of growth in program purchases at list price is due to increased utilization, while only 20% is due to drug price increases. Provider-driven factors that increase utilization include ongoing provider consolidation due to practice acquisition, growth of contract pharmacy networks including the establishment of in-house specialty pharmacies, a higher propensity for 340B providers to treat patients using drugs versus their non-340B peers [...], and ambiguity in how some covered entities interpret patient eligibility rules.

— IQVIA, “The Size and Growth of the 340B Program in 2025,” Introduction

During the life of the 340B program, two mechanisms have been used to deliver lower cost drugs — upfront discounts and rebates. With an upfront discount, the 340B hospital or clinic pays a reduced price at the time of purchase, lowering its initial acquisition cost. In contrast, under a rebate model, the covered entity initially pays a non-340B price at the time of purchase. That price may be the full list price or WAC, a lower Group Purchasing Organization (GPO) discounted price, or another price. Whatever the initial price, a rebate is then used to reduce the 340B purchase price to the 340B price. While both upfront and rebate mechanisms reduce the cost of drugs, they differ in timing, which may affect the covered entity's cash flow.

— IQVIA, “How Will a Rebate Model Impact Cash Flow in the 340B Drug Pricing Program?” (Sun, Zeng, Sarraile, Martin), Discount mechanisms

Reflection B1. List every stakeholder named or implied in the three excerpts above (there are at least six). For each, write one sentence on whether they prefer an upfront discount or a rebate, and why.

Payers and market access

Rare disease products are often perceived as extremely high price, with headlines referencing million-dollar price tags per patient all too often a highly visible aspect of rare disease product launch. These headlines, of course, obscure the fact that the overall cost of rare disease treatments to medicines budgets is in fact quite small at 11% of total prescription spending in both the U.S. and Europe. While per patient costs can be high, the number of patients is small [...]. Healthcare systems are not well set up to cover high upfront payments for curative one-off gene therapies costing \$2 million or more per patient. [...] A mixture of long-term trends (payer tightening of budgets), and new policies aimed at cost containment in the aftermath of the pandemic are making market access more challenging generally.

— IQVIA, “From Orphan to Opportunity: Mastering Rare Disease Launch Excellence” (Lutzmayr, Scott, Rickwood), A toughening market access environment

Note the reframing: the same drug is “a million-dollar price tag” to a payer, “11% of spend” to a manufacturer, and a once-in-a-lifetime treatment to a patient. Analysts are constantly asked to translate between these framings — with the same underlying numbers.

Providers: what an HCP actually wants from the industry

As artificial intelligence accelerates access to scientific information within clinical workflows, HCPs are increasingly navigating a landscape where data is abundant, but interpretation is complex. The differentiator is no longer access to data but the ability to interpret, contextualise, and apply evidence in real-world settings, grounded in human judgement, contextual understanding, and trust. [...] “88% of HCPs search for medical information daily or weekly; over 50% already use AI tools.” [...] HCPs need scientific partners who can help them make sense of evidence within the realities of their clinical practice and not just share information, but contextualise it, challenge it where needed, and apply it meaningfully to patient pathways and decisions.

— IQVIA, “Redefining Global Medical Scientific Engagement” (Evans), Executive summary

Reflection B2. The excerpt describes medical (MSL) engagement. How does a sales rep's objective differ from an MSL's, and why does U.S. law keep them separate? (Hint: look up “promotional” vs. “non-promotional” in the glossary.)

Patients as consumers: when a prescription category reshapes a market

Obesity is no longer a peripheral public-health issue or a niche commercial category. In many affluent markets, a majority of adults are already overweight or living with obesity. The arrival of highly effective GLP-1 and GIP/GLP-1 medicines changes the commercial baseline because clinically meaningful weight loss is now possible at scale, at least for people who can access, tolerate and remain on treatment. [...] The decisive constraints are no longer efficacy alone. They are affordability, reimbursement, supply, tolerability, persistence, maintenance after weight loss and the service infrastructure needed to support long-term treatment.

— IQVIA, “If Better Weight-Loss Drugs Keep Improving, Consumer Health Will Be Reshaped” (Rickwood & Spitzer), Executive summary

Every constraint in that last sentence maps to a stakeholder and to an analytics problem in this course: affordability and reimbursement (payers, Week 6 access scenarios), supply (wholesalers, forecasting), tolerability and persistence (patients, adherence models in Week 5), service infrastructure (hubs and patient services).

C. The manufacturer's view: AI, data and the commercial P&L (objectives 1 & 3)

The last three excerpts describe the world you are being trained to enter: what commercial leaders are spending on AI, why their projects stall, and how they decide where to deploy generative and agentic AI.

Survey findings show that AI is moving rapidly from experimentation to execution. Over 80% of organizations have advanced beyond pilots, and more than a third now describe themselves as “AI Advanced.” [...] Nearly half of the surveyed companies dedicate more than 20% of their commercial budgets to AI, with another quarter investing between 11% and 20%. [...] 58% of leaders report achieving 2X ROI from their AI initiatives within a single year, with 7% seeing 3X or more. However, scaling AI — and realizing its full value — remains challenging. Data privacy, legacy systems, and talent shortages are the most common barriers. Notably, 36% of organizations say their data is either “mostly insufficient” or only “moderately sufficient” to support AI, highlighting a persistent gap in data readiness. Just over half (52%) rate their data as “mostly sufficient,” and only 11% say it is “fully sufficient.”

— IQVIA, “AI in Life Sciences Commercialization” (Kotakonda & Awasthi), Executive summary — May 2025 survey of 107 senior commercial leaders

Sales and marketing are rated highest in strategic importance, yet low current average adoption. Despite the promise of AI to drive field productivity, omnichannel targeting, and content personalization, many organizations have not yet industrialized these tools. The barriers are less about ambition and more about scaling enablement across decentralized teams. [...] Value and access along with Compliance also carry priority but show even lesser adoption. AI use cases frequently cited in these areas include payer analytics, audit automation, and pricing simulation. [...] While organizations use varied data sources — such as CRM, field activity, claims, prescriptions, digital engagement, patient support, and market data — these are often siloed and inconsistently structured, limiting their utility for enterprise-scale AI. As one respondent put it, “We have the data — we just can't use it the way we need to.”

— IQVIA, “AI in Life Sciences Commercialization,” AI impact by commercial function; Barriers to scale

Too many pilots and proofs-of-concept fail to work as intended. Across industries, more than 70% of GenAI experiments fail to make it to production. [...] [Four major reasons programs fail:] 1. Unrealistic expectations that GenAI can create deep insights from limited data — LLMs are unable to create insights if the information necessary for those insights isn't somewhere in the data sources. 2. Implementations don't fit within users' workflow — Inconvenience kills value. 3. Challenges to prove required accuracy — errors in GenAI are less predictable and harder to understand when validating a new service. 4. Not all use cases are useful for end users — this is especially true for expert users who have become fast and effective at using systems.

— IQVIA, “Where to Deploy Generative AI Solutions in Life Sciences” (Rink & O'Reilly), Introduction; Core reasons why GenAI applications sometimes fail

LLMs can rarely do things a person can't do if given enough time. The vast majority of successful GenAI products automate human-like tasks vs. doing something completely novel. [...] A process that has a bigger impact on the P&L will have more business value. For example, a tool to evaluate companies for M&A may get less frequent use but have more value than an AI tool that generates new PowerPoint templates. [...] Agentic AI, which is empowered to perform dedicated tasks autonomously, is expected to dramatically augment what GenAI is capable of.

— IQVIA, “Where to Deploy Generative AI Solutions in Life Sciences,” Business impact; Understanding Generative AI

Reflection C. Pick one line of the GLUCOREN brand P&L from tonight's slides (e.g., gross-to-net deductions, field-force spend, DTC). Propose one AI use case that would move that line, then score it on IQVIA's two axes — business impact, and cost-and-risk to implement. Would it be a “quick win,” a “major project,” a “fill-in” or a “time task”?

D. The official reading and what to look for

Reading	What to read	What to extract
IQVIA Institute — “The Use of Medicines in the U.S.” (latest annual report)	Executive summary (≈8 pages); skim the charts on spending, net vs. list price, and utilization by therapy area	(1) the gap between invoice/list spending and net spending; (2) which therapy areas drive growth; (3) how much patients pay out of pocket. Bring one number that surprised you.
Course glossary (Week_01_Course_Glossary)	All of it, once	Be able to explain the ten “test yourself” terms without notes.
Optional: FDA — “Framework for FDA's Real-World Evidence Program” (2018)	Executive summary and Section II	What counts as RWD vs. RWE; why the distinction matters for Week 2.

Source list

- IQVIA. A Framework for Successful Biopharma Launches (Gupta, M.; McClellan, W.).
- IQVIA. Inflection Point: How Clinical Trial Results Impact Biopharma Valuations (Gores, M.; Harries, W.; Capuzzi, F.).
- IQVIA. The Size and Growth of the 340B Program in 2025 (Martin, R.; Karne, H.; Zeng, S.), 2026.
- IQVIA. How Will a Rebate Model Impact Cash Flow in the 340B Drug Pricing Program? (Sun, C.; Zeng, S.; Sarraile, W.; Martin, R.), 2025.
- IQVIA. From Orphan to Opportunity: Mastering Rare Disease Launch Excellence (Lutzmayer, S.; Scott, K.; Rickwood, S.).
- IQVIA. Redefining Global Medical Scientific Engagement: Operationalising Human Centric Excellence in an AI Accelerated World (Evans, C.).

- IQVIA Consumer Health. If Better Weight-Loss Drugs Keep Improving, Consumer Health Will Be Reshaped (Rickwood, S.; Spitzer, V.).
- IQVIA. AI in Life Sciences Commercialization: Strategic Insights and Practical Recommendations from 2025 Survey of Commercial Leaders (Kotakonda, S.; Awasthi, P.), 2025.
- IQVIA. Where to Deploy Generative AI Solutions in Life Sciences: Learning from Common Pitfalls in Commercial Generative AI Efforts (Rink, C.; O'Reilly, W.), 2025.

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